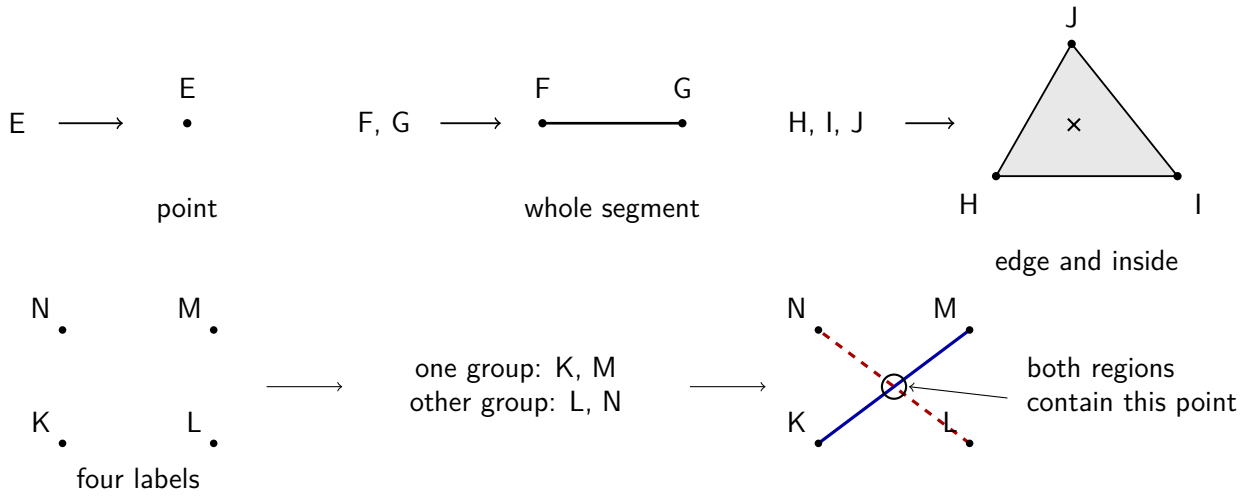
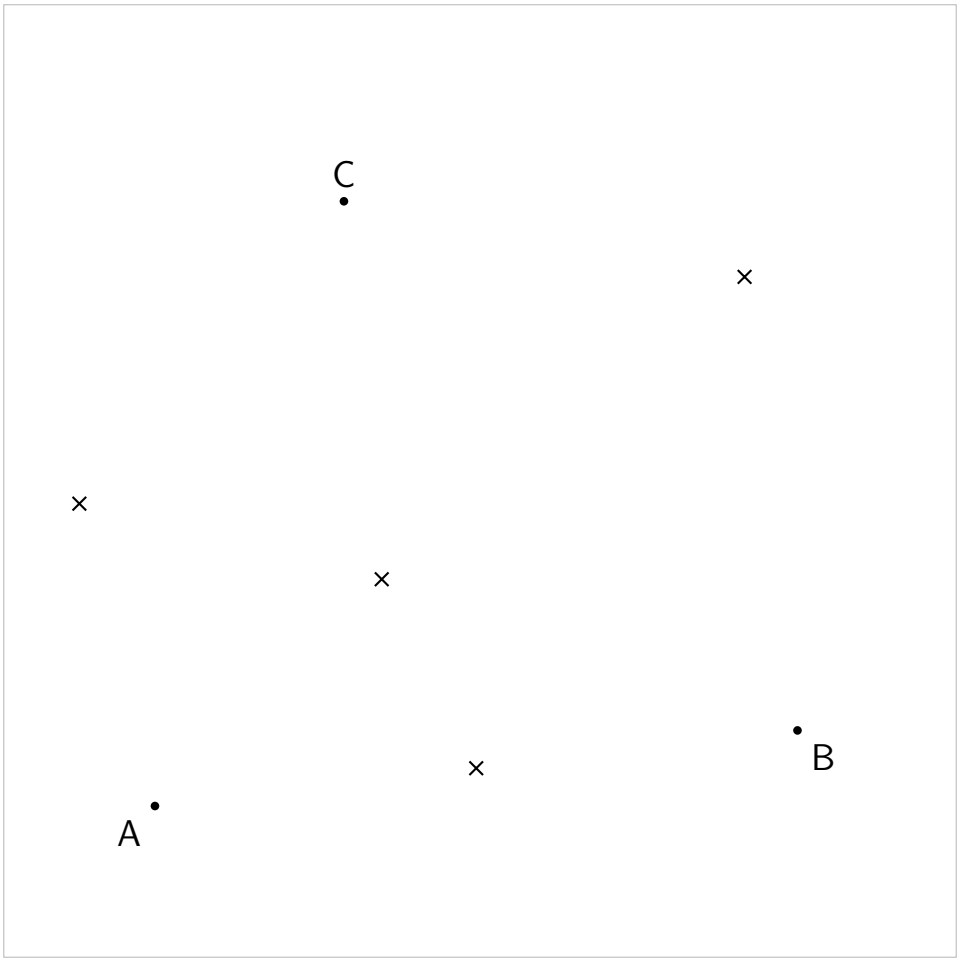
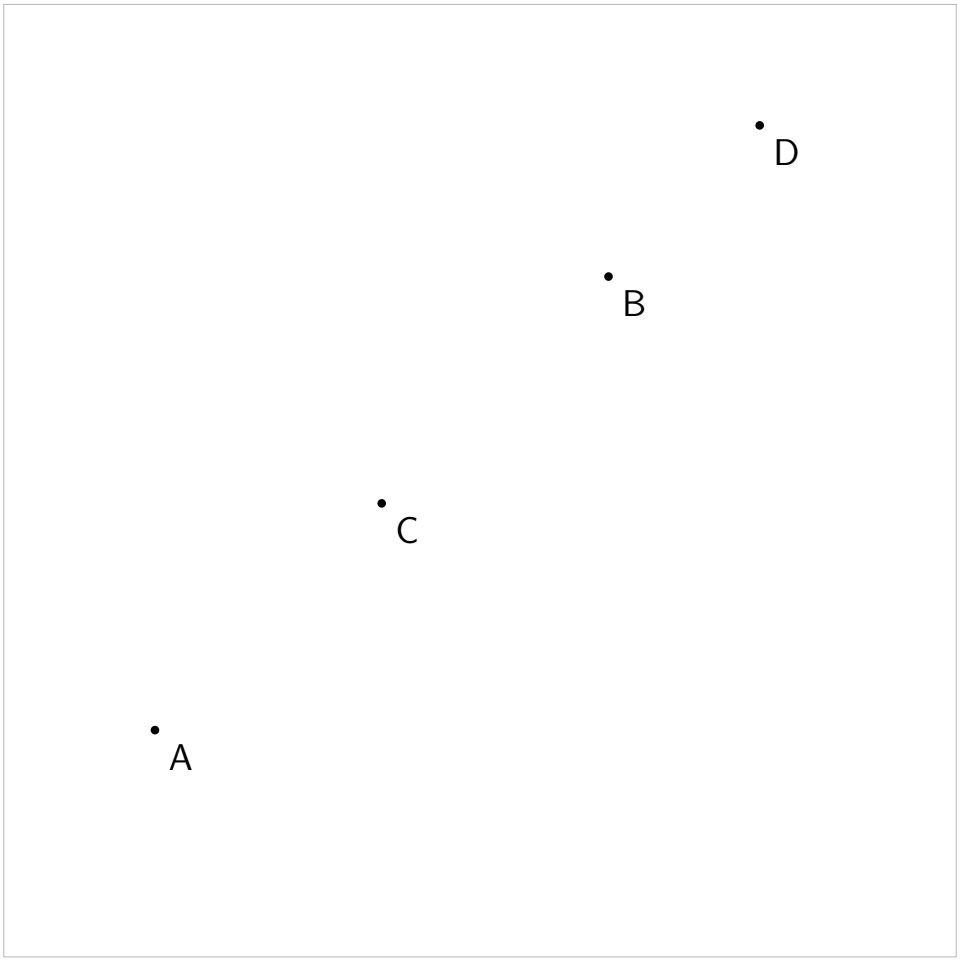
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split.



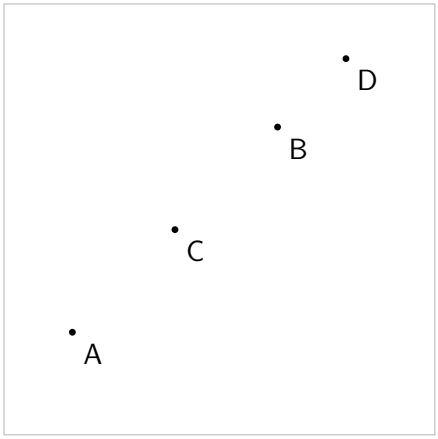
Problem 1: Put D on each cross. For each position, split A, B, C, and D so their two regions meet, and mark a shared point.

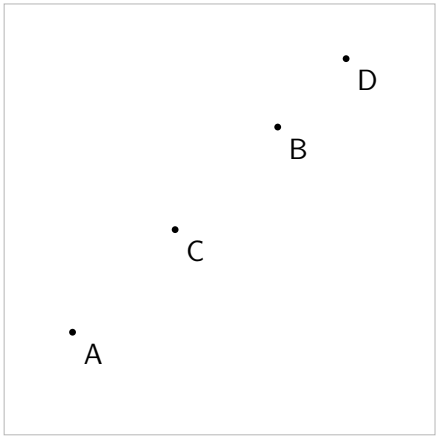


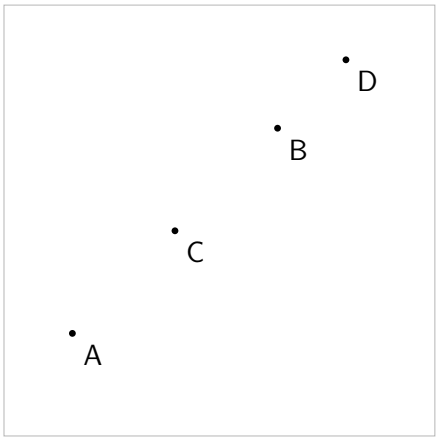
Problem 2: Find every split whose regions meet. Explain how you know your list is complete.

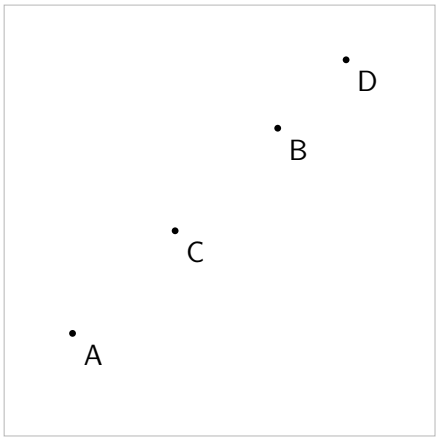


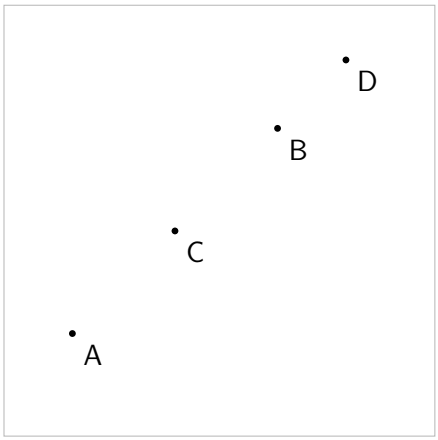
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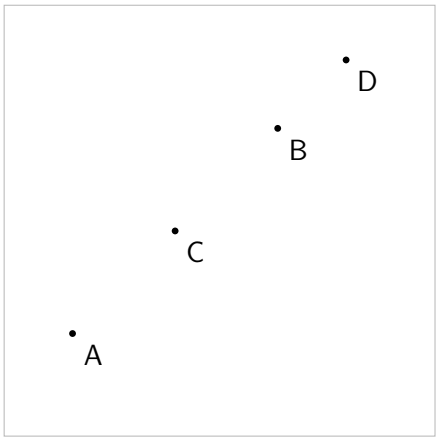






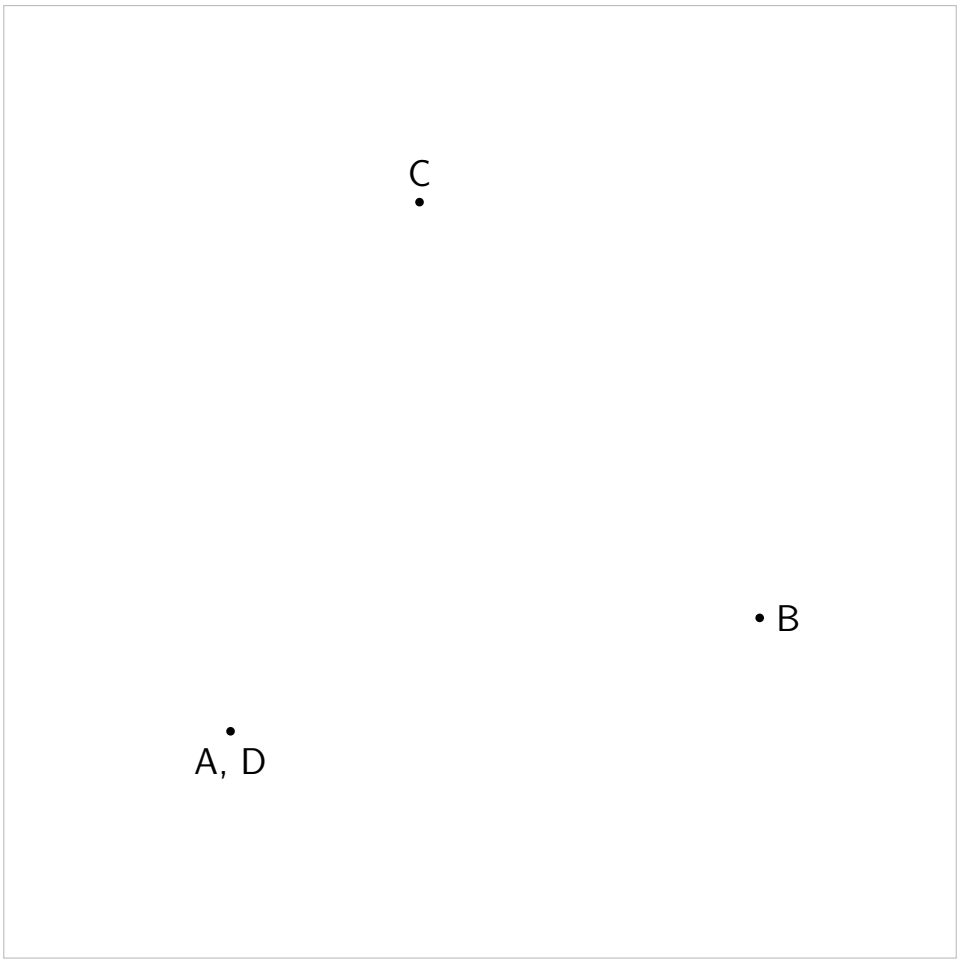




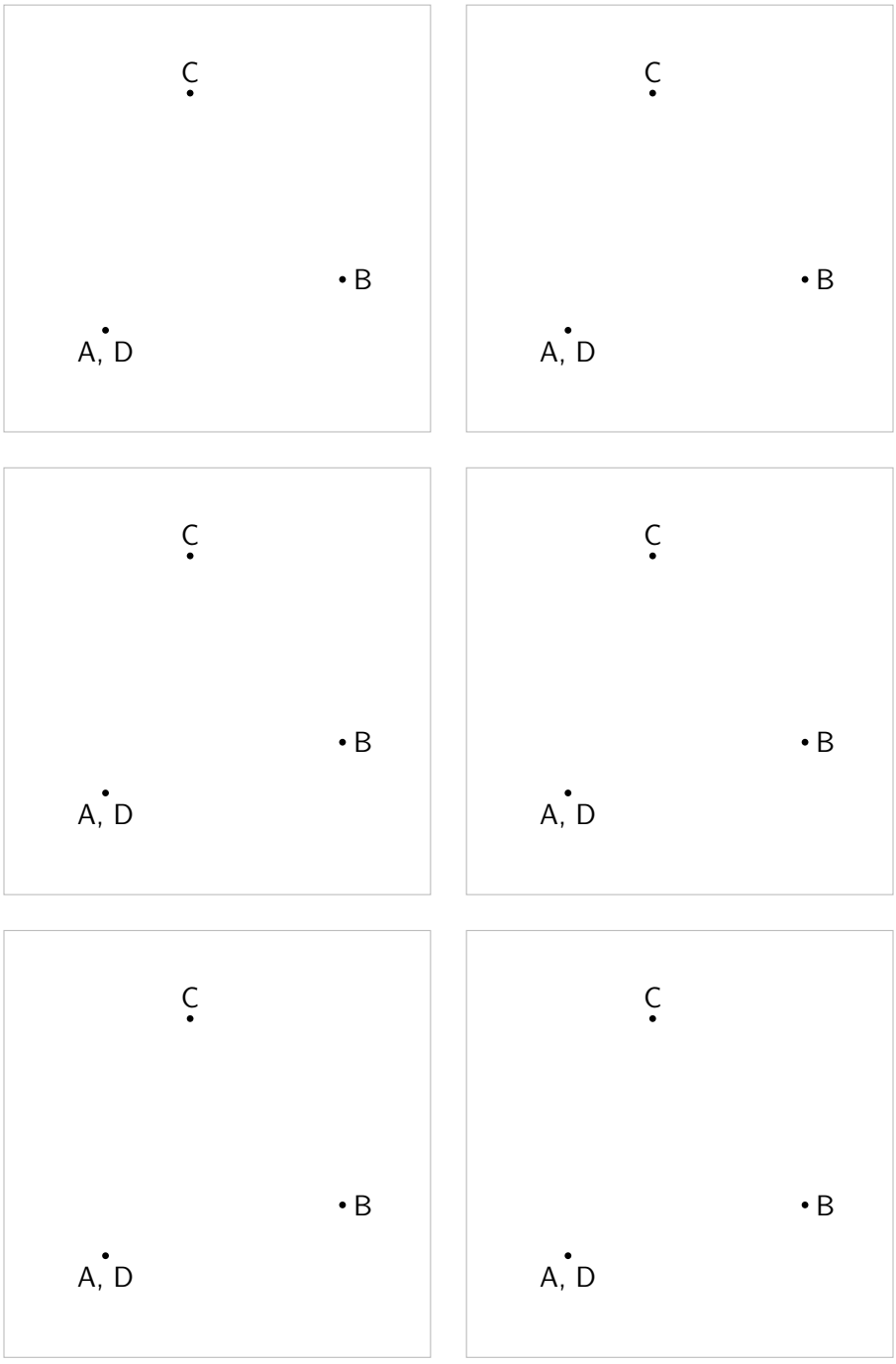


Problem 3: Take turns placing A, B, C, and D in the box. Try to make an arrangement with no successful split. Decide whether your opponent must always be able to find one.

Problem 4: A and D share a location. Find every split whose regions meet. Does a successful split always exist when two of four labels share a location? Explain.



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Problem 5: Make two four-dot arrangements that each have exactly one successful split, with different successful splits. Make a third arrangement with more than one successful split. Explain how you checked the number of successful splits.

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Problem 6: Can every arrangement of A, B, C, and D be split into two groups whose regions meet? Give an explanation that covers all arrangements, including dots in a straight line and labels at the same point.

Problem 7: Can these three dots be split into two groups whose regions meet? Explain why your answer covers every split. Find the smallest number of dots that guarantees a successful split for every arrangement, and explain why fewer cannot give that guarantee.

